

Experiment 1 — Exploratory Data Analysis (EDA)

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Aim

To perform Exploratory Data Analysis (EDA) on the Aadhar Enrollment dataset to understand its structure, detect and treat missing values, and summarize key statistical properties using descriptive measures such as mean, standard deviation, min, max, and quartiles.

Theory

Exploratory Data Analysis (EDA) is the process of systematically examining a dataset before applying any formal modeling or statistical inference. It helps in understanding the nature of data, finding patterns, spotting anomalies, and checking assumptions.

1. Data Loading and Preview: The dataset is loaded into a pandas DataFrame from an external file (Excel/CSV). The head() function is used to display the first few rows, giving an initial overview of the data — its columns, values, and general structure.

2. Dataset Dimensions: Understanding the size of the dataset is fundamental. The shape attribute returns the number of rows (records) and columns (features). For the Aadhar dataset, rows represent individual enrollment records and columns represent attributes such as demographic details, registration status, and enrollment statistics.

3. Descriptive Statistics: The describe() function generates a statistical summary for all numeric columns. Key measures include:

- **Mean** — Average value of a column, representing central tendency.
- **Standard Deviation** — Measures how spread out the values are from the mean.
- **Min / Max** — The smallest and largest values observed in a column.
- **Quartiles (25%, 50%, 75%)** — Divide the data into four equal parts; the 50th percentile is the median.

These statistics give a quick sense of the scale, range, and distribution of each numeric attribute.

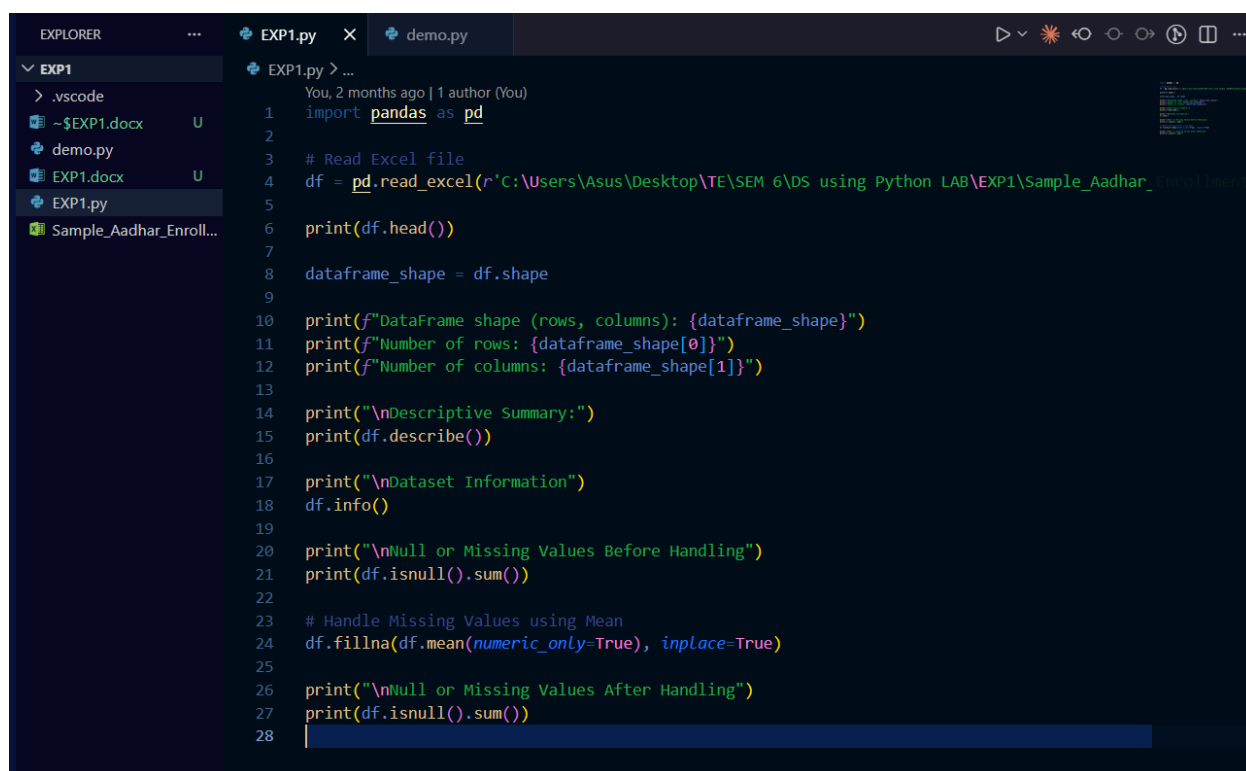
4. Data Type Information: The `info()` function provides column-wise details including data types (int, float, object), non-null counts, and memory usage. This helps identify columns that may need type conversion or special handling.

5. Missing Value Detection: Missing values (NaN) are common in real-world datasets and can arise from incomplete data entry, system errors, or unavailable records. The `isnull().sum()` function returns the count of missing values per column, helping identify which columns require treatment.

6. Missing Value Treatment (Imputation): Once missing values are detected, they are filled using the `fillna()` method. For numeric columns, the mean of the respective column is used for imputation. Mean imputation is a standard technique that preserves the overall distribution without drastically altering summary statistics.

7. Verification: After imputation, `isnull().sum()` is run again to confirm that no missing values remain in the dataset, ensuring data quality before any further analysis.

Execution



```
EXPLORER  ...  EXP1.py  X  demo.py
EXP1
  > .vscode
  ~$EXP1.docx  U
  demo.py
  EXP1.docx  U
  EXP1.py
  Sample_Aadhar_Enroll...

EXP1.py > ...
You, 2 months ago | 1 author (You)
1  import pandas as pd
2
3  # Read Excel file
4  df = pd.read_excel(r'C:\Users\Asus\Desktop\TE\SEM 6\DS using Python LAB\EXP1\Sample_Aadhar_Enrollment.xlsx')
5
6  print(df.head())
7
8  dataframe_shape = df.shape
9
10 print(f'DataFrame shape (rows, columns): {dataframe_shape}')
11 print(f'Number of rows: {dataframe_shape[0]}')
12 print(f'Number of columns: {dataframe_shape[1]}')
13
14 print('\nDescriptive Summary:')
15 print(df.describe())
16
17 print('\nDataset Information')
18 df.info()
19
20 print('\nNull or Missing Values Before Handling')
21 print(df.isnull().sum())
22
23 # Handle Missing Values using Mean
24 df.fillna(df.mean(numeric_only=True), inplace=True)
25
26 print('\nNull or Missing Values After Handling')
27 print(df.isnull().sum())
28
```

Output

```

EXPLORER      ...  PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS
EXP1
> .vscode
~$EXP1.docx  U
demo.py
EXP1.docx  U
EXP1.py
Sample_Aadhar_Enroll...

PS C:\Users\Asus\Desktop\TE\SEM 6\DS using Python LAB\EXP1> python EXP1.py
      Date      State      district      pincode      age_0_5      age_5_17      age_18_greater
0  2025-02-03 00:00:00  Meghalaya  East Khasi Hills  793121.0      11.0      61.0      37.0
1  2025-09-03 00:00:00  Karnataka  Bengaluru Urban  560043.0      14.0      33.0      39.0
2  2025-09-03 00:00:00  Uttar Pradesh  Kanpur Nagar  208001.0      29.0      82.0      12.0
3  2025-09-03 00:00:00  Uttar Pradesh  Aligarh  202133.0      62.0      29.0      15.0
4  2025-09-03 00:00:00  Karnataka  Bengaluru Urban  560016.0      14.0      16.0      21.0

DataFrame shape (rows, columns): (99, 7)
Number of rows: 99
Number of columns: 7

Descriptive Summary:
      pincode      age_0_5      age_5_17      age_18_greater
count      96.000000      94.000000      94.000000      95.000000
mean    530493.010417      29.808511      49.117021      21.821053
std    266900.060688      23.710746      44.224466      20.114025
min    110018.000000      10.000000      10.000000      10.000000
25%    265808.000000      14.250000      16.250000      13.000000
50%    560035.500000      20.000000      33.500000      17.000000
75%    783570.000000      35.000000      59.500000      23.500000
max    847421.000000     134.000000     249.000000     148.000000

Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99 entries, 0 to 98
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Date        99 non-null    object
1   State       98 non-null    object
2   district    94 non-null    object
3   pincode     96 non-null    float64
4   age_0_5     94 non-null    float64
5   age_5_17    94 non-null    float64
6   age_18_greater  95 non-null    float64
dtypes: float64(4), object(3)
    
```

```

EXPLORER      ...  PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS
EXP1
> .vscode
~$EXP1.docx  U
demo.py
EXP1.docx  U
EXP1.py
Sample_Aadhar_Enroll...

PS C:\Users\Asus\Desktop\TE\SEM 6\DS using Python LAB\EXP1> python EXP1.py

Null or Missing Values Before Handling
Date      0
State      1
district    5
pincode     3
age_0_5     5
age_5_17    5
age_18_greater  4
dtype: int64

Null or Missing Values After Handling
Date      0
State      1
district    5
pincode     0
age_0_5     0
age_5_17    0
age_18_greater  0
dtype: int64

PS C:\Users\Asus\Desktop\TE\SEM 6\DS using Python LAB\EXP1>
    
```

Conclusion

In this experiment, Exploratory Data Analysis (EDA) was performed on the Aadhar Enrollment dataset. The dataset structure, data types, and statistics were analyzed using functions like `head()`, `shape`, `info()`, and `describe()`. Missing values were identified and handled by replacing them with the mean of numeric columns using `fillna()`. After cleaning, the dataset became free of null values and ready for further analysis or machine learning tasks.